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Department of Computer Science

Faculty of Science and Technology

CSC2210: OBJECT ORIENTED PROGRAMMING 2

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Section: [P]

Group No: 11

Project Report On “Water & Sewage Management System using C#”

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Submitted By -

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CO2: Display and verify the mean of a real-life Project using the concepts of C# Graphical User Interface based environment with database integration to depict a desktop-based application.

Assessment Criteria	Not Attended/ Incorrect (0)	Inadequate (1-2)	Average (3)	Good (4)	Excellent (5)
Evaluation Criteria	Evaluation Definition				Total
Requirement fulfillment	Properly demonstrate a real-life scenario-based project with proper functional requirement identification for the Object-Oriented Programming project development activities.				
Validation	Ensuring the ability of students’ proper demonstration on validation forms in their system in terms of dealing with the data.				
Verification	Identifying if the students can verify the system data along with proper functional requirements in terms of data flow.				

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1. INTRODUCTION

This smart billing, complaint, connection and maintenance management system is a role-based desktop application developed using C# windows Forms and Microsoft SQL Server. The project mainly focuses on common water service operations such as user registration, bill generation, bill dispute handling, complaint submission, complaint assignment, connection application processing, notice publication and maintenance task tracking.

The codebase is organized into separate folders for Forms and Resources. This simple structure helps the team separate user interface codes, database code and business logic.

2. PROBLEM STATEMENT

Manual management of water services can create billing mistakes, slow compliant handling, poor application tracking, weak communication, and difficulty in monitoring field-level maintenance work. Customers often need to contact the office repeatedly to know bill status, complaint progress, or connection application updates. The proposed software solves these administrative problems by centralizing records and providing separate dashboards for each actor.

3. OBJECTIVE OF THE SYSTEM

- To develop a desktop based OOP2 project using C# Windows Forms.
- To use Microsoft SQL Server to store users, customers, bills, complaints, connection applications, notices, payments, maintenance tasks and reports.
- To provide role-based Login and redirect each user to the correct dashboards.
- To allow Admin to approve employee accounts, manage users, publish notices, assign complaints, and view reports.
- To Allow Service Officers to generate bills, review bill disputes, verify connection applications, and produce billing reports.
- Allow Customers to view bills, submit disputes, submit complaints, apply for connections, and track status
- Allow Maintenance Engineers to view assigned tasks, update repair progress, set visit dates, write inspection notes, and submit completion reports.
- Demonstrate OOP concepts such as classes, objects, encapsulation, separation of responsibilities, and reusable helper methods.

4. TOOLS & TECHNOLOGIES

Technology/Tool	Usage in Project
C# Language	Programming language used for application logic.
Windows Forms	User interface framework for desktop forms and dashboards
.NET Framework 4.7.2	Runtime framework used by the Visual Studio project.
Microsoft SQL Server	Relational database used to store project records.
Git & GitHub	Used for team collaboration, branch work, and code upload.

5. ACTORS & FEATURE LIST

a. Admin Features

- Login
- Logout
- Register
- View Profile
- Update Profile
- Approve Pending Employees
- Manage Users
- Assign Complaints
- Manage Notices
- Check System Reports
- Change Password
- Forget Password

b. Customer Features

- Login
- Logout
- Register

- View Profile
- Update Profile
- View Current Bill
- View Bill History
- Submit Bill Dispute
- Submit Complaint
- Track Complaint Status
- Apply for New Connection
- Track Connection Application
- Change Password
- Forget Password

c. Service Officer Features

- Login
- Logout
- Register
- View Profile
- Update Profile
- Enter Meter Reading
- Generate Bill
- View Customer Bill List
- Review Disputes
- Correct Bill Amount
- Review Connection Application
- Verify Documents
- Schedule Installations
- Generate Billing Report
- Change Password
- Forget Password

d. Maintenance Engineer

- Login
- Logout
- Register

- View Profile
- Update Profile
- View Assigned Complaints
- View Maintenance Tasks
- Update Repair Progress
- Add Inspection Notes
- Set Expected Visit Date
- Submit Completion Report
- Change Password
- Forget Password

6. DATABASE DESIGN

Table Name	Attributes
BillDisputes	DisputeID, BillID, CustomerID, Reason, Status, ReviewedBy, SubmittedAt
Bills	BillID, CustomerID, BillingMonth, PreviousReading, CurrentReading, UnitUsed, Amount, Arrears, Status, CreatedAt, GeneratedBy
Complaints	ComplaintID, CustomerID, Category, Description, Priority, Status, AssignedEngineerID, DateSubmitted
ConnectionApplications	ApplicationID, CustomerID, ApplicationDate, DocumentStatus, ApprovalStatus, RejectionReason, AssignedOfficer, InstallationDate
Customers	CustomerID, UserID, MeterNumber, HoldingNumber, ConnectionType
MaintenanceTasks	TaskID, ComplaintID, EngineerID, VisitDate, ProgressStatus, Notes, CompletionReport, UpdatedAt
Notices	NoticeID, Title, Description, Area, NoticeType, PublishedBy, PublishDate
Payments	PaymentID, BillID, CustomerID, PaymentDate, Amount, Method, ReceiptNo

Reports	ReportID, CreatedBy, ReportType, CreatedDate, Description
Users	UserID, FullName, Email, Phone, Password, Role, Address, Status, CreatedAt

Lenovo\SQL...BillDisputes			
	Column Name	Data Type	Allow Nulls
▶	DisputeID	int	☐
	BillID	int	☐
	CustomerID	int	☐
	Reason	varchar(500)	☐
	Status	varchar(30)	☒
	ReviewedBy	int	☒
	SubmittedAt	datetime	☒
			☐

Figure 6.1 - BillDispute Table

Lenovo\SQL... - dbo.Bills			
	Column Name	Data Type	Allow Nulls
▶	BillID	int	☐
	CustomerID	int	☐
	BillingMonth	varchar(30)	☐
	PreviousReading	int	☒
	CurrentReading	int	☒
	UnitUsed		☒
	Amount	decimal(10, 2)	☐
	Arrears	decimal(10, 2)	☒
	Status	varchar(30)	☒
	CreatedAt	datetime	☒
	GeneratedBy	int	☒
			☐

Figure 6.2 - Bills Table

Lenovo\SQL...Complaints			
	Column Name	Data Type	Allow Nulls
▶	ComplaintID	int	☐
	CustomerID	int	☐
	Category	varchar(50)	☐
	Description	varchar(500)	☐
	Priority	varchar(20)	☒
	Status	varchar(30)	☒
	AssignedEngineerID	int	☒
	DateSubmitted	datetime	☒
			☐

Figure 6.3 - Complaints Table

Lenovo\SQL...Applications			
	Column Name	Data Type	Allow Nulls
▶	ApplicationID	int	☐
	CustomerID	int	☐
	ApplicationDate	datetime	☒
	DocumentStatus	varchar(30)	☒
	ApprovalStatus	varchar(30)	☒
	RejectionReason	varchar(300)	☒
	AssignedOfficer	int	☒
	InstallationDate	date	☒
			☐

Figure 6.4 - ConnectionApplication Table

Lenovo\SQL...o.Customers			
	Column Name	Data Type	Allow Nulls
▶	CustomerID	int	☐
	UserID	int	☐
	MeterNumber	int	☒
	HoldingNumber	varchar(50)	☒
	ConnectionType	varchar(50)	☒
			☐

Figure 6.5 - Customer Table

Lenovo\SQL...enanceTasks			
	Column Name	Data Type	Allow Nulls
▶	TaskID	int	☐
	ComplaintID	int	☐
	EngineerID	int	☐
	VisitDate	date	☒
	ProgressStatus	varchar(30)	☒
	Notes	varchar(500)	☒
	CompletionReport	varchar(500)	☒
	UpdatedAt	datetime	☒
			☐

Figure 6.6 - MaintananceTasks Table

Lenovo\SQL...dbo.Notices			
	Column Name	Data Type	Allow Nulls
▶	NoticeID	int	☐
	Title	varchar(100)	☐
	Description	varchar(500)	☐
	Area	varchar(100)	☒
	NoticeType	varchar(50)	☒
	PublishedBy	int	☐
	PublishDate	datetime	☒
			☐

Figure 6.7 - Notice Table

Lenovo\SQL...bo.Payments			
	Column Name	Data Type	Allow Nulls
▶	PaymentID	int	☐
	BillID	int	☐
	CustomerID	int	☐
	PaymentDate	datetime	☒
	Amount	decimal(10, 2)	☐
	Method	varchar(30)	☒
	ReceiptNo	varchar(50)	☒
			☐

Figure 6.8 - Payment Table

Lenovo\SQL...dbo.Reports			
	Column Name	Data Type	Allow Nulls
PK	ReportID	int	☐
	CreatedBy	int	☐
	ReportType	varchar(50)	☐
	CreatedDate	datetime	☒
	Description	varchar(500)	☒
			☐

Figure 6.9 - Report Table

Lenovo\SQL...- dbo.Users			
	Column Name	Data Type	Allow Nulls
PK	UserID	int	☐
	FullName	varchar(100)	☐
	Email	varchar(100)	☐
	Phone	varchar(20)	☒
	Password	varchar(100)	☐
	Role	varchar(30)	☐
	Address	varchar(200)	☒
	Status	varchar(20)	☒
	CreatedAt	datetime	☒
	Gender	nvarchar(10)	☒
	DOB	date	☒
			☐

Figure 6.10 - Users Table

7. ER DIAGRAM

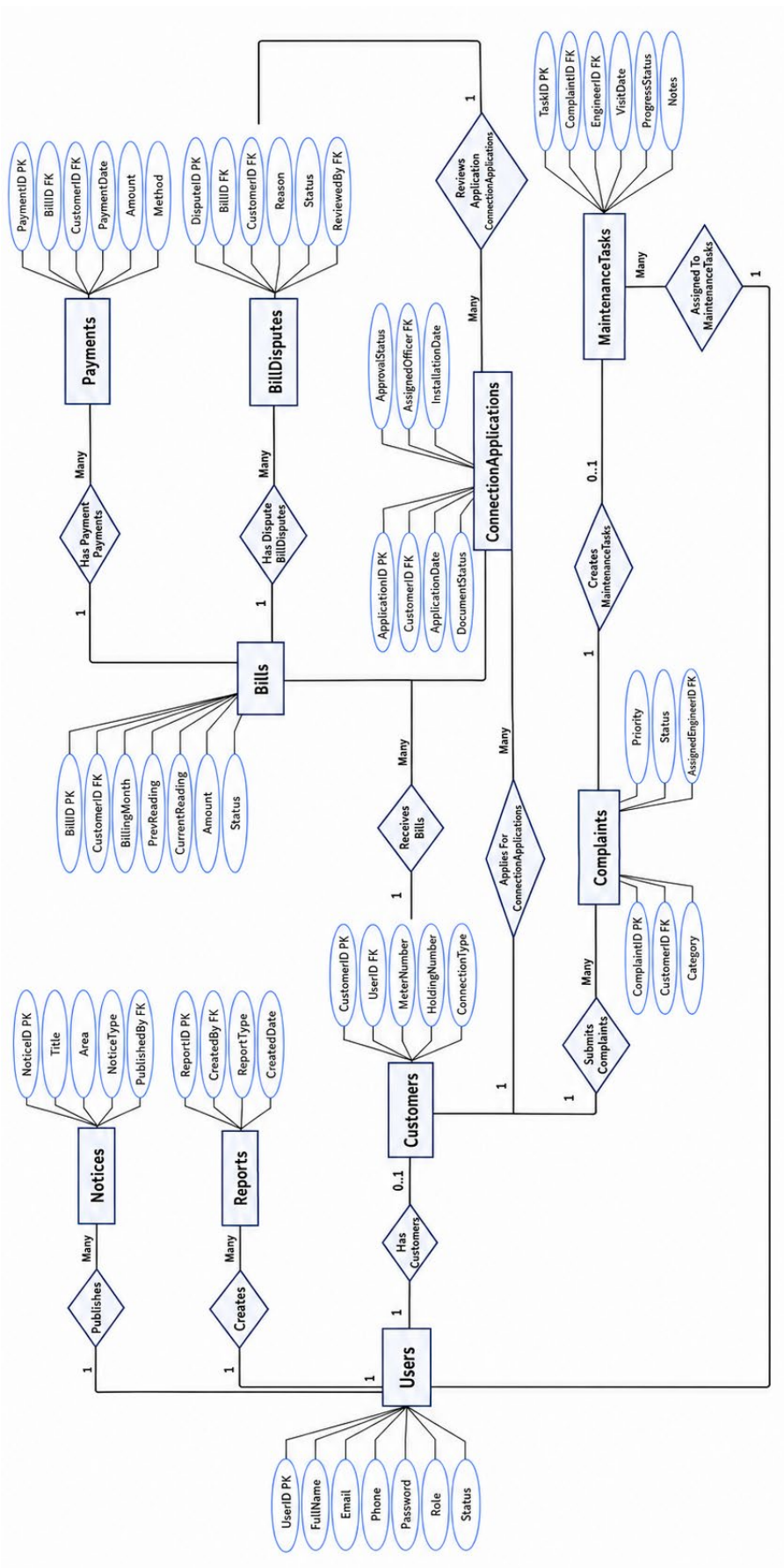


Figure 7.1 - ER Diagram for the Project

8. NORMALIZED FORM OF THE TABLE

i. Has Customers

3NF:

- **Users:** UserID, FullName, Email, Phone, Password, Role, Status
- **Customers:** CustomerID, UserID, MeterNumber, HoldingNumber, ConnectionType

ii. Publishes Notices

3NF:

- **Users:** UserID, FullName, Email, Phone, Password, Role, Status
- **Notices:** NoticeID, Title, Area, NoticeType, PublishedBy

iii. Creates Reports

3NF:

- **Users:** UserID, FullName, Email, Phone, Password, Role, Status
- **Reports:** ReportID, CreatedBy, ReportType, CreatedDate

iv. Receives Bills

3NF:

- **Customers:** CustomerID, UserID, MeterNumber, HoldingNumber, ConnectionType
- **Bills:** BillID, CustomerID, BillingMonth, PrevReading, CurrentReading, Amount, Status

v. Has Payment Payments

3NF:

- **Bills:** BillID, CustomerID, BillingMonth, PrevReading, CurrentReading, Amount, Status
- **Payments:** PaymentID, BillID, CustomerID, PaymentDate, Amount, Method

vi. Has Dispute BillDisputes

3NF:

- **Bills:** BillID, CustomerID, BillingMonth, PrevReading, CurrentReading, Amount,
- **BillDisputes:** DisputeID, BillID, CustomerID, Reason, Status, ReviewedBy
- **Users:** UserID, FullName, Email, Phone, Password, Role, Status

vii. Applies For ConnectionApplications

3NF:

- **Customers:** CustomerID, UserID, MeterNumber, HoldingNumber, ConnectionType
- **ConnectionApplications:** ApplicationID, CustomerID, ApplicationDate, DocumentStatus, ApprovalStatus, AssignedOfficer, InstallationDate
- **Users:** UserID, FullName, Email, Phone, Password, Role, Status

viii. Submits Complaints

3NF:

- **Customers:** CustomerID, UserID, MeterNumber, HoldingNumber, ConnectionType
- **Complaints:** ComplaintID, CustomerID, Category, Priority, Status, AssignedEngineerID
- **Users:** UserID, FullName, Email, Phone, Password, Role, Status

ix. Creates MaintenanceTasks

3NF:

- **Complaints:** ComplaintID, CustomerID, Category, Priority, Status, AssignedEngineerID
- **MaintenanceTasks:** TaskID, ComplaintID, EngineerID, VisitDate, ProgressStatus, Notes
- **Users:** UserID, FullName, Email, Phone, Password, Role, Status

x. Assigned To MaintenanceTasks

3NF:

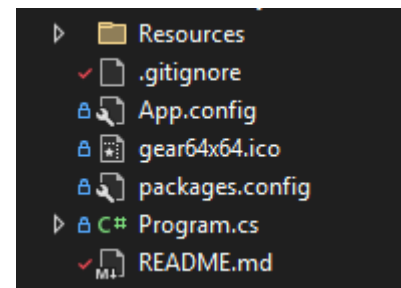
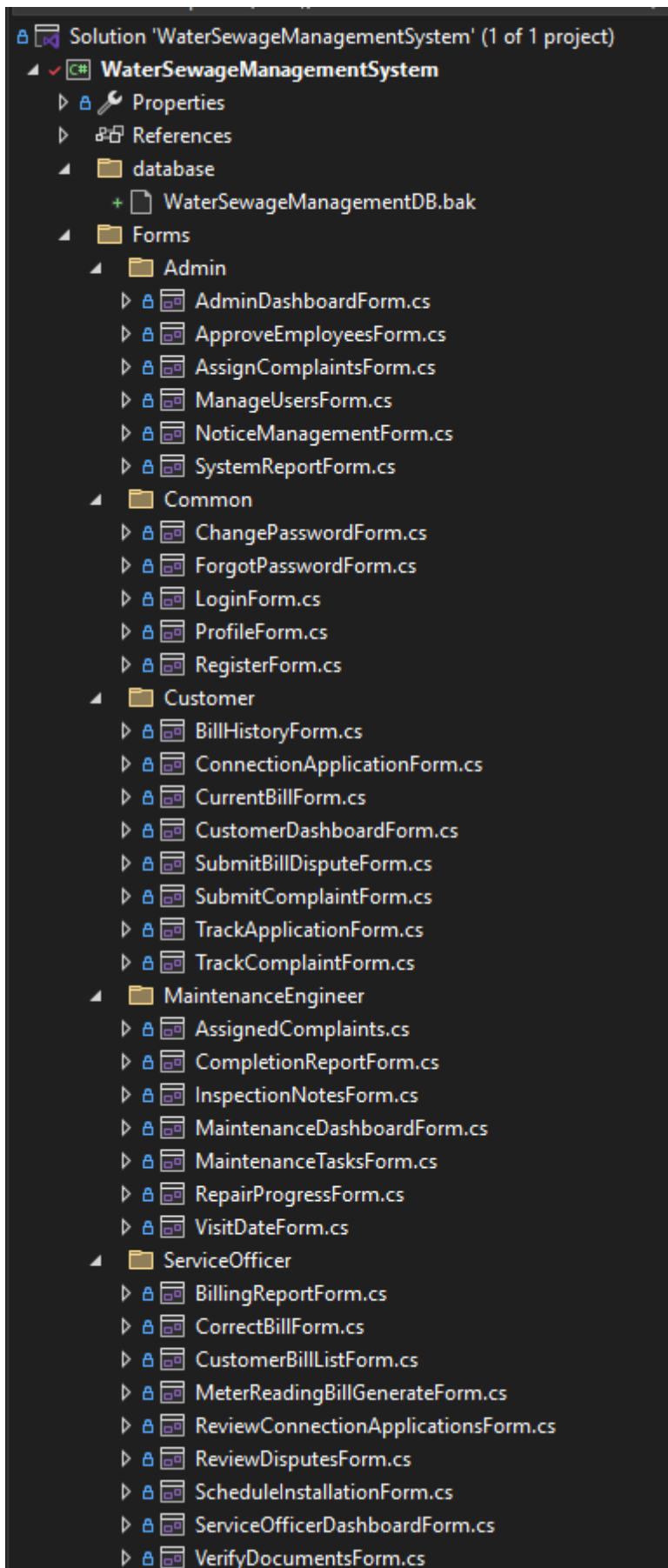
- **Users:** UserID, FullName, Email, Phone, Password, Role, Status
- **MaintenanceTasks:** TaskID, ComplaintID, EngineerID, VisitDate, ProgressStatus, Notes

Finalization

After normalization up to 3NF, the final tables are:

- i. **Users:** UserID, FullName, Email, Phone, Password, Role, Status
- ii. **Customers:** CustomerID, UserID, MeterNumber, HoldingNumber, ConnectionType
- iii. **Bills:** BillID, CustomerID, BillingMonth, PrevReading, CurrentReading, Amount, Status
- iv. **Payments:** PaymentID, BillID, CustomerID, PaymentDate, Amount, Method
- v. **BillDisputes:** DisputeID, BillID, CustomerID, Reason, Status, ReviewedBy
- vi. **Complaints:** ComplaintID, CustomerID, Category, Priority, Status, AssignedEngineerID
- vii. **MaintenanceTasks:** TaskID, ComplaintID, EngineerID, VisitDate, ProgressStatus, Notes
- viii. **ConnectionApplications:** ApplicationID, CustomerID, ApplicationDate, DocumentStatus, ApprovalStatus, AssignedOfficer, InstallationDate
- ix. **Notices:** NoticeID, Title, Area, NoticeType, PublishedBy
- x. **Reports:** ReportID, CreatedBy, ReportType, CreatedDate

9. PROJECT FOLDER STRUCTURE



10. Sample of CRUD Operations

Create:

```
string query = @"INSERT INTO Notices
                (Title, Description, Area, NoticeType, PublishedBy, PublishDate)
                VALUES
                (@Title, @Description, @Area, @NoticeType, @PublishedBy, GETDATE())";

SqlCommand cmd = new SqlCommand(query, conn);
cmd.Parameters.AddWithValue("@Title", title);
cmd.Parameters.AddWithValue("@Description", description);
cmd.Parameters.AddWithValue("@Area", area);
cmd.Parameters.AddWithValue("@NoticeType", noticeType);
cmd.Parameters.AddWithValue("@PublishedBy", publishedBy);
```

Read:

```
string queryMeter = @"SELECT COUNT(*)
                      FROM Bills
                      WHERE PreviousReading IS NOT NULL
                      AND CurrentReading IS NOT NULL
                      AND GeneratedBy = @OfficerID";

SqlCommand cmdMeter = new SqlCommand(queryMeter, conn);
cmdMeter.Parameters.AddWithValue("@OfficerID", officerID);
lblMeterCount.Text = cmdMeter.ExecuteScalar().ToString();
```

Update:

```
string query = "UPDATE MaintenanceTasks SET Notes='" + notes +
               "', UpdatedAt=GETDATE() WHERE TaskID=" + taskID;

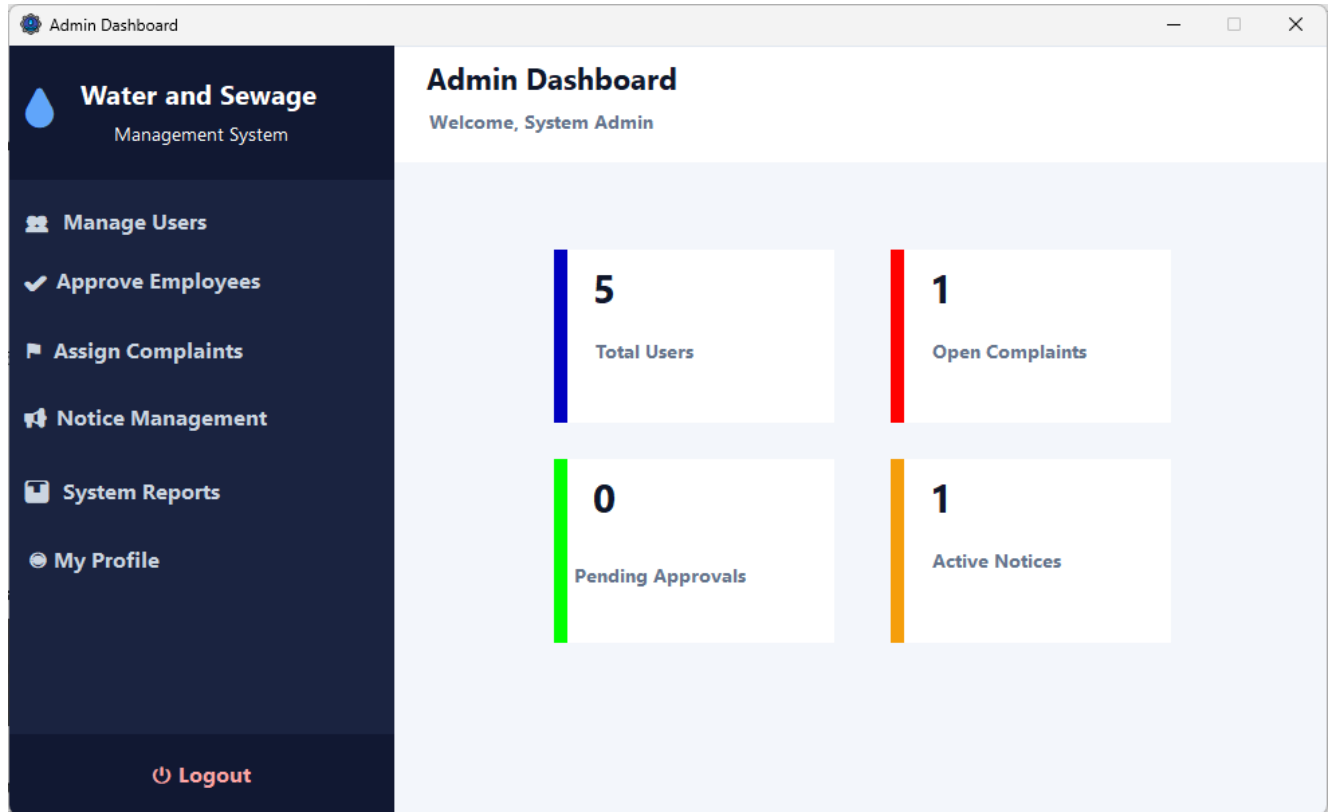
SqlCommand cmd = new SqlCommand(query, conn);
int rows = cmd.ExecuteNonQuery();
```

Delete:

```
string deleteUserQuery = "DELETE FROM Users WHERE UserID = @UserID";
SqlCommand userCmd = new SqlCommand(deleteUserQuery, conn);
userCmd.Parameters.AddWithValue("@UserID", currentUserID);
```


11. Application Form Screenshots

a. Admin Forms



The Manage Users interface displays a table of system accounts. The table has the following columns: UserID, FullName, Email, Phone, Role, Address, Gender, Status, and CreatedAt. Below the table are buttons for Activate, Deactivate, Refresh, and Close.

UserID	FullName	Email	Phone	Role	Address	Gender	Status	CreatedAt
7	test	test@gmail.com	04204800202	Customer	dhaka	Male	Active	20-May-26 2:14...
1	System Admin	admin@gmail.c...	01700000001	Admin	Dhaka	Male	Active	13-May-26 8:16...
2	Customer One	customer@gma...	01700000002	Customer	Mirpur	Male	Active	13-May-26 8:16...
3	Service Officer ...	service@gmail...	01700000003	ServiceOfficer	WASA Office	Male	Active	13-May-26 8:16...
4	Maintenance En...	engineer@gmai...	01700000004	MaintenanceEn...	Field Office	Male	Active	13-May-26 8:16...

Approve Employee

Approve Employees

Review and approve pending employee registrations

UserID	FullName	Email	Phone	Role	Address	Gender	Status	CreatedAt
--------	----------	-------	-------	------	---------	--------	--------	-----------

Approve Selected

Refresh

Close

Assign Complaints

Assign Complaints

Select a complaint, pick an engineer, then assign

ComplaintID	CustomerID	CustomerName	Category	Description	Priority	Status	AssignedEngine	EngineerName	DateSubmitted
1	1	Customer One	Water Supply	No water supply ...	High	Resolved	4	Maintenance En...	19-May-26 10:3...

Assign to Engineer:

Maintenance Engineer One

Assign

Refresh

Close



Notice Management

Publish new notices and manage existing ones

Title *

Description *

Area

Type

General

NoticeID	Title	Description	Area	NoticeType	PublishedBy	PublishedByName	PublishDate
1	Water Supply Mana...	Water Supply may b...	Kuratoli	Maintenance	1	System Admin	19-May-26 10:44 PM

Publish Notice

Delete Selected

Close

Publish new notices and manage existing ones

Title *

--

Description *

--

Area

--

Type

General

NoticeID	Title	Description	Area	NoticeType	PublishedBy	PublishedByName	PublishDate
1	Water Supply Mana...	Water Supply may b...	Kuratoli	Maintenance	1	System Admin	19-May-26 10:44 PM

Publish Notice

Delete Selected

Close

Log and review system activity reports

Report Type:

Description:

Log Report

ReportID	CreatedBy	CreatedByName	ReportType	CreatedDate	Description
3	3	Service Officer Asif	Billing	19-May-26 10:46 PM	Service Officer generated bilin...
2	1	System Admin	Complaint	19-May-26 10:46 PM	Complaint workflow tested suc...
1	4	Maintenance Engineer One	Maintenance	19-May-26 10:42 PM	Completion report submitted for...

Close

b. Customer Forms

Customer Dashboard

Current Bill

Bill History

Submit Complaint

Track Complaint

+ New Connection

Track Application

My Profile

Logout

Water and Sewage Management System - Customer

Welcome, Customer

0

Current Bill

2

Bill History

0

Open Complaints

Approved

Connection Status

View Notice

Notice ID	Title	Description	Area	Notice Type	Published By	Publish Date
1	Water Supply...	Water Supply may be interu...	Kuratoli	Maintenance	1	19 May 2026

Current Bill

Your Current Bill

Bill #2

Month: May 2026

Previous: 50 | Current: 80 | Units Used: 30

Bill Amount: Tk. 100.00

Arrears: Tk. 0.00

Total Due: Tk. 100.00

Status: Paid

Submit Dispute

Pay Bill

Close

Bill History

Bill History

BillID	Customer	Customer	MeterNo	BillingM	Previous	Current	Amount	Arrears	TotalDue	Status	Created
2	1	Custom...	101	May 2026	50	80	100.00	0.00	100.00	Paid	20-May-...
1	1	Custom...	101	May 2026	0	50	300.00	0.00	300.00	Paid	19-May-...

Refresh

Close

Submit Complaint

Submit Complaint

Category

Priority

Description

Submit

Cancel

Track Complaint

Track Complaint

ComplaintID	CustomerID	Category	Description	Priority	Status	DateSubmitted
1	1	Water Supply	No water supply...	High	Resolved	19-May-26 10:3...

Refresh

Close

Apply for New Connection

+ New Connection

Submitting this application will request a new water/sewage connection.

Step 1: Submit this application form

Step 2: Service Officer will verify your documents

Step 3: Application will be approved or rejected

Step 4: Installation will be scheduled if approved

NID Number *

Connection Type *

Submit Application

Close

Track Application

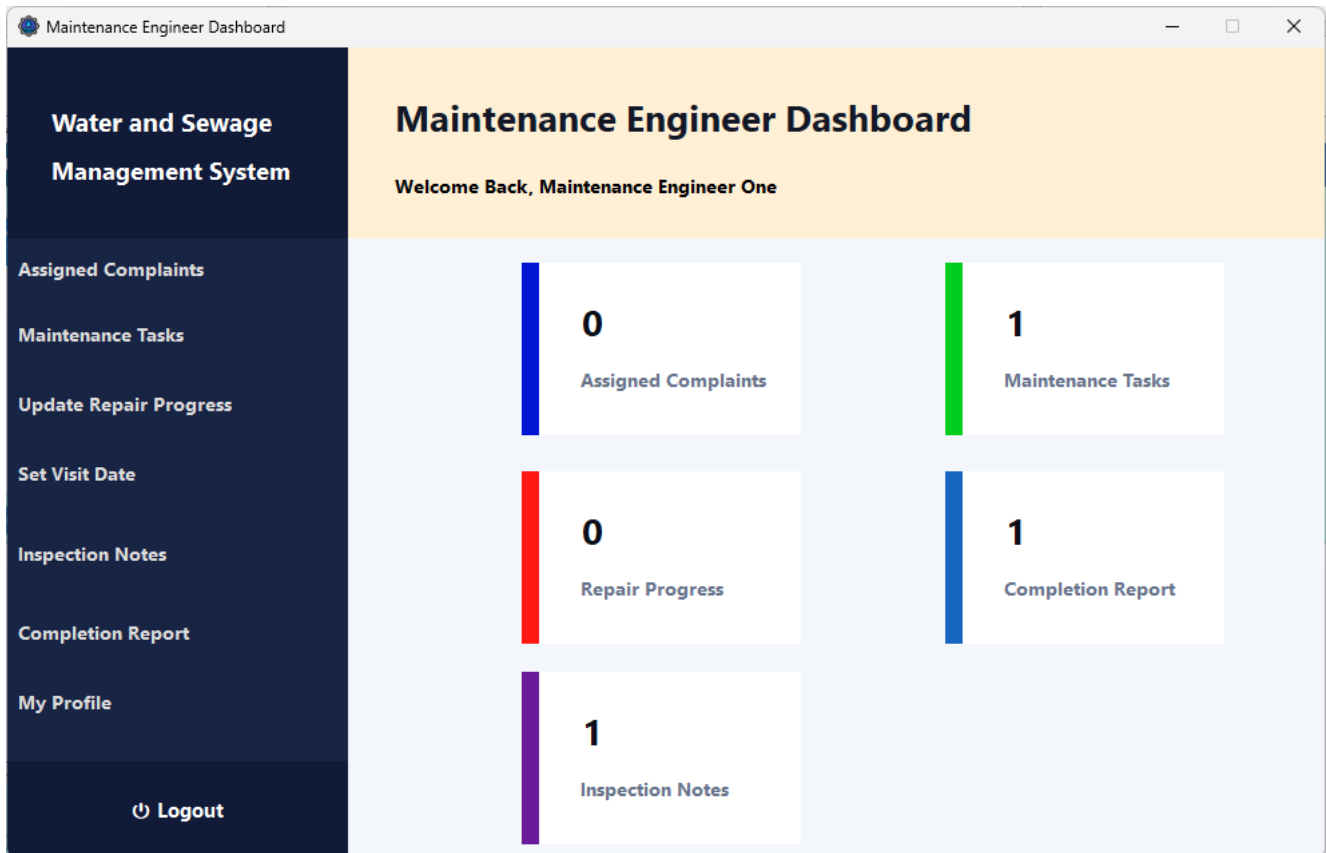
Track Application

Appli...	Custo...	Custo...	Holdi...	Meter...	Conn...	Appli...	Docu...	Appro...	Reject...	Assig...	Install...
1	1	Custom...	KHA-101	101	Reside...	19-May...	Verified	Approved		3	21-May...

Refresh

Close

c. Maintenance Engineer Form



Assigned Complaints								
ComplaintID	CustomerID	CustomerName	Category	Description	Priority	Status	AssignedEngineerID	Da
1	1	Customer One	Water Supply	No water supply in my area since morning	High	Resolved	4	19

Mark In Progress

Mark Resolved

Refresh

Close

My Maintenance Tasks

Maintenance Tasks

TaskID	ComplaintID	EngineerID	VisitDate	ProgressStatus	Notes	CompletionReport	Update
1	1	4	21 May 2026	Completed	Completed the task	The issue was solved after valve adjustments. Supply restored	19 May 2

RefreshClose

Update Repair Progress

Update Repair Progress

TaskID	ComplaintID	EngineerID	VisitDate	ProgressStatus	Notes	CompletionReport	Update
1	1	4	21 May 2026	Completed	Completed the task	The issue was solved after valve adjustments. Supply restored	19 May 2

New Status:

Notes:

Update ProgressClearClose

Set Expected Visit Date

Set Expected Visit Date

Select a task from the list, then choose the date you plan to visit.

TaskID	ComplaintID	EngineerID	VisitDate	ProgressStatus	Notes	CompletionReport	Update
1	1	4	21 May 2026	Completed	Completed the task	The issue was solved after valve adjustments. Supply restored	19 May 2

Visit Date:

20-May-26

Set Visit Date

Refresh

Close

Field Inspection Notes

Field Inspection Notes

Click a task row to load its current notes, edit them, then click Save Notes.

TaskID	ComplaintID	EngineerID	VisitDate	ProgressStatus	Notes	CompletionReport	Update
1	1	4	21 May 2026	Completed	Completed the task	The issue was solved after valve adjustments. Supply restored	19 May 2

Inspection Notes:

Completed the task

Save Notes

Clear

Close

Maintenance Completion Report

Select a task, write your completion report, then click Submit. This will mark the task as Completed.

TaskID	ComplaintID	EngineerID	VisitDate	ProgressStatus	Notes	CompletionReport	Update
1	1	4	21 May 2026	Completed	Completed the task	The issue was solved after valve adjustments. Supply restored	19 May 2

Completion Report:

The issue was solved after valve adjustments. Supply restored

Submit Completion Report

Clear

Close

d. Service Officer Forms

Service Officer Dashboard

Water and Sewage Management System - Service Officer

Welcome, Service Officer Asif

Service Officer Dashboard

- Meter Read and Bill Generate
- Customer Bill List
- Review Disputes
- Verify Documents
- Review Applications
- Schedule Meter Installation
- Log Billing Report
- My Profile

Logout

2	0
Meter Readings	Pending Disputes
2	0
Paid Bills	Connection Requests

Enter Meter Reading & Generate Bill

Meter Reading and Bill Generation

Customer : Customer One - Meter: 101 - Resider ▾

Billing Month : May 2026

Previous Reading : 80

Current Reading :

Arrears : 0

Generate Bill **Close**

Customer Bill List

List of Customer Bills

BillID	CustomerID	CustomerName	MeterNumber	BillingMonth	PreviousReading	CurrentReading	Amount	Arrears	Status	CreatedAt
2	1	Customer One	101	May 2026	50	80	100.00	0.00	Paid	20-May-26 12:31 ...
1	1	Customer One	101	May 2026	0	50	300.00	0.00	Paid	19-May-26 10:28 ...

Mark as Paid

Close

Review Bill Disputes

Review Bill Disputes

DisputeID	BillID	CustomerID	CustomerName	BillingMonth	BillAmount	Reason	Status	ReviewedBy	SubmittedAt
2	2	1	Customer One	May 2026	100.00	Too much for me	Reviewed	3	20-May-26 12:31 AM
1	1	1	Customer One	May 2026	300.00	The bill amount see...	Reviewed	3	19-May-26 10:29 PM

Mark as Reviewed

Correct Bill

Close

Document Verification

Documents to review

ApplicationID	CustomerID	CustomerName	HoldingNumbe	MeterNumber	ConnectionTyp	ApplicationDat	DocumentStat	ApprovalStatus	RejectionReas	AssignedOffice	InstallationDat
2	3	test	1234	102	Commercial	20-May-26 2:15...	Verified	Approved		3	
1	1	Customer One	KHA-101	101	Residential	19-May-26 10:2...	Verified	Approved		3	21-May-26

Verify Documents

Close

Review Connection Application

List of Connection Applications

ApplicationID	CustomerID	CustomerName	HoldingNumbe	MeterNumber	ConnectionTyp	ApplicationDat	DocumentStat	ApprovalStatus	RejectionReas	AssignedOffice	InstallationDat
2	3	test	1234	102	Commercial	20-May-26 2:15...	Verified	Approved		3	
1	1	Customer One	KHA-101	101	Residential	19-May-26 10:2...	Verified	Approved		3	21-May-26

Approve

Reject

Meter Number -

Close

Schedule Installation

Schedule Installation

ApplicationID	CustomerID	CustomerName	HoldingNumber	MeterNumber	ConnectionType	ApplicationDate	DocumentStatus	ApprovalStatus	RejectionReason	AssignedOffice	InstallationDate
2	3	test	1234	102	Commercial	20-May-26 2:15...	Verified	Approved		3	
1	1	Customer One	KHA-101	101	Residential	19-May-26 10:2...	Verified	Approved		3	21-May-26

Installation Date:

21-May-26

Schedule

Close

Billing Reports

Billing Reports

BillID	CustomerID	CustomerName	MeterNumber	BillingMonth	PreviousReading	CurrentReading	UnitUsed	Amount	Arrears	Status	CreatedAt
2	1	Customer One	101	May 2026	50	80	30	100.00	0.00	Paid	20-May-26 12:3...
1	1	Customer One	101	May 2026	0	50	50	300.00	0.00	Paid	19-May-26 10:2...

Log Report

Close

e. Common Forms

Water & Sewage Management System - Login

Water and Sewage Managment System

Sign in to your account

Email Address

Password

☐ Show Password

LOGIN

[Don't have an account? Register](#)

[Forgot Password?](#)



Register New Account

Create Account

Full Name *

Date of Birth *

Email *

Password *

Phone *

Confirm Password *

Gender *

☐ Male ☐ Female

Register As *

Address

REGISTER

Cancel

 Reset Password

Reset Password

Registered Email

New Password

Confirm Password

Reset Password

Cancel

 My Profile

My Profile

Role: Customer
Status: Active

Full Name

Email

Phone

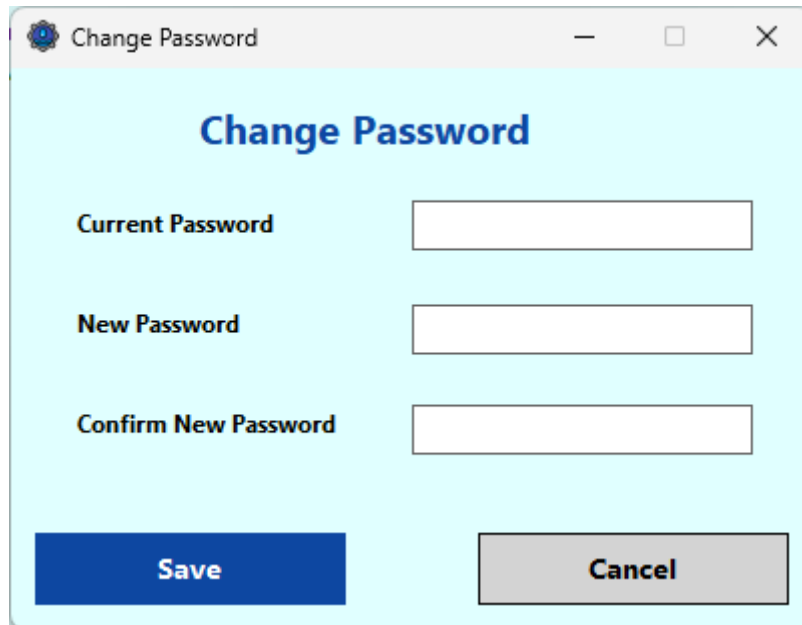
Address

Save Profile

Change Password

Delete Profile

Close

A screenshot of a 'Change Password' dialog box. The window has a title bar with a gear icon, the text 'Change Password', and standard minimize, maximize, and close buttons. The main area has a light blue background with the title 'Change Password' in bold blue text. Below the title are three input fields: 'Current Password', 'New Password', and 'Confirm New Password'. At the bottom are two buttons: a blue 'Save' button and a grey 'Cancel' button.

Change Password

Current Password

New Password

Confirm New Password

Save Cancel

12. Limitations and Future Improvement

a. Limitations

- The application is a desktop-based academic project, so it is not currently available online.
- Payment gateway integration is not implemented in the current version.
- SMS/email notifications are not implemented in the current version.
- The system improves administrative tracking but cannot directly solve physical water supply or pipeline infrastructure problems.
- Some reports and dashboard counters may need final polishing before presentation.

b. Future Improvements

- Add online payment support for water bills.
- Add SMS or email notification for bill generation, complaint assignment, and connection status updates.
- Add map/location support for complaints.
- Add more dashboard charts for monthly bills, complaints, and maintenance statistics.
- Create a backup/restore feature for the database.

- Improve UI navigation by loading forms inside dashboard panels instead of opening separate popup windows.

13. Conclusion

The Smart WASA Billing, Complaint, Connection and Maintenance Management System is a practical C# Windows Forms application for an OOP2 final project. It includes four clear actors, a relational SQL Server database, role-based login, billing, complaints, connection applications, maintenance tasks, notices, and reports. The project is not too simple because it contains multiple real-life modules, but it is also manageable for a four-member team within a short development period.

Overall, the system demonstrates important Object Oriented Programming concepts and provides a realistic software solution for managing administrative water and sewage service operations.

14. References

- **GitHub Repository:** <https://github.com/Noble-Knight-7/Water-and-Sewage-Management-System>
- **ER Diagram:** <https://app.diagrams.net/>